Name: ______________________

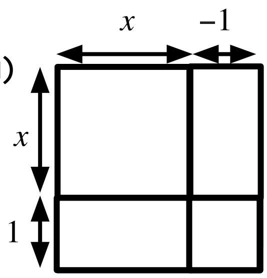
Worksheet

Difference of two squares

Task A: Multiplying binomials

1) Fill in the area models for these products of two binomials. Write your answer as a simplified expression

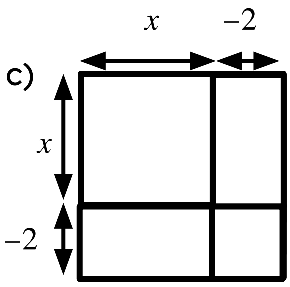
a)



$(x + 1)\ (x - 1)$

______

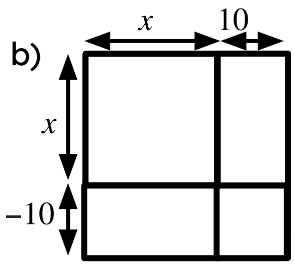
c)



$(x - 2)\ (x - 2)$

______

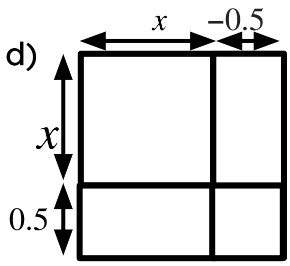
b)



$(x - 10)\ (x + 10)$

______

d)



$(x + 0.5)\ (x - 0.5)$

______

2) Which of these products of two binomials can be simplified to an expression with exactly two terms?

a) $(x + 9)(x - 8)$

b) $(x - 9)(x - 9)$

c) $(x - 6)(x + 5)$

d) $(x - 13)(x + 13)$

e) $(x + 7)(x + 7)$

f) $(x + 11)(x - 11)$

g) $(x - 2.4)(x + 2.4)$

h) $(x - 5)(x + 5.5)$

i) $(x + \frac{11}{2})(x - 5.5)$

j) $(x - 0.8)(x + \frac{4}{5})$

k) $(x + \frac{3}{8})(x - 0.37)$

l) $(x - \frac{17}{20})(x - 1.85)$

3) Match a binomial in the left column with a binomial in the middle column and find its product in the end column.

a)	$(x - 6)$	i)	$(x - 9)$	q)	$x^2 - 49$
b)	$(x - 5)$	j)	$(x - 7)$	r)	$x^2 + 5x - 36$
c)	$(x - 2)$	k)	$(x - 5)$	s)	$x^2 - 4$
d)	$(x + 1)$	l)	$(x - 4)$	t)	$x^2 - 1$
e)	$(x + 2)$	m)	$(x - 2)$	u)	$x^2 - 10x + 25$
f)	$(x + 4)$	n)	$(x - 1)$	v)	$x^2 - x - 2$
g)	$(x + 7)$	o)	$(x + 1)$	w)	$x^2 - 5x - 36$
h)	$(x + 9)$	p)	$(x + 6)$	x)	$x^2 - 36$

Task B: Identifying difference of two squares

1) Which of these products of two binomials can be written as the difference of two squares?

a) $(x + 1)(x - 4)$

e) $(x - y)(x + y)$

i) $(xy - 3)(xy + 3)$

b) $(y + 1)(x - 1)$

f) $(3a - b)(3a - b)$

j) $(3x - 2y)(-3x + 2y)$

c) $(a - 46)(a + 46)$

g) $(10 + x)(10 - x)$

2) Fill in the area models for these products of two binomials. Write your answer as the difference of two squares.

a)

y^2	
	-400

$(y + 20)(y - 20)$

$\equiv$ ______

c)

	$-4y^2$

$(x - 2y)(x + 2y)$

$\equiv$ ______

b)

	$10x$
$-10x$	

$(x - 10)(x + 10)$

$\equiv$ ______

d)

$(a - 3b)(a + 3b)$

$\equiv$ ______

e)

9	$-3x$

$(3 + x)(3 - x)$

$\equiv$ ______

g)

x^2	
	-16

$(x -$ ______$)(x +$ ______$)$

$\equiv x^2 - 16$

f)

$(2y - 3x)(2y + 3x)$

$\equiv$ ______

h)

$25a^2$	$15ab$
	$-9b^2$

(______ + ______) (______ – ______)

$\equiv$ (______)^2 – (______)^2

$\equiv 25a^2 - 9b^2$